

MathMind AI – RAG Test Document 2

This document is designed to test PDF ingestion, chunking, vector retrieval, and question answering in the MathMind AI RAG Assistant.

1. Algebra

A linear equation has the form $ax + b = 0$, where a is not zero. For example, $3x + 6 = 0$ gives $x = -2$. A quadratic equation has the form $ax^2 + bx + c = 0$, where a is not zero. Its discriminant is $b^2 - 4ac$.

2. Geometry

The area of a rectangle is length multiplied by width. The area of a triangle is one-half times base times height. For a circle, the area is πr^2 and the circumference is $2\pi r$.

3. Statistics

The mean is calculated by adding all observations and dividing by the number of observations. The median is the middle value after sorting the observations. The mode is the value that occurs most frequently.

4. Probability

For equally likely outcomes, probability of an event is the number of favorable outcomes divided by the total number of possible outcomes. A probability is always between 0 and 1 inclusive.

5. Worked Example

Suppose a rectangle has length 12 cm and width 5 cm. Its area is $12 \times 5 = 60 \text{ cm}^2$ and its perimeter is $2(12 + 5) = 34 \text{ cm}$.

6. RAG Test Questions

Question	Expected topic
What is the area of the rectangle in the worked example?	Geometry
What is the discriminant of $ax^2 + bx + c$?	Algebra
How is the median calculated?	Statistics
What is the range of a valid probability?	Probability
What is the perimeter of the example rectangle?	Worked example